

MONASH University

Are you in the room? A framework for sortition and representation

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A thesis submitted for the degree of
Bachelor of Commerce (Honours)
at Monash University in 2026
Department of Econometrics & Business Statistics

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Abstract

Sortition, the random selection of individuals to form panels, has seen a sort of renaissance in recent years, as more and more institutions realise the value of direct community and stakeholder input when making decisions. These panels have proven to be massively beneficial in terms of improving the quality of decisions, increasing public trust and engagement, and creating more inclusive and representative decision-making processes. However, there are still many challenges to overcome in order to ensure that these panels are truly representative and effective. This thesis aims to evaluate current sortition methods and community representation processes, as well as to investigate how unobserved traits and survey non-response may affect panel representativeness. By investigating these factors, we can develop a greater understanding of existing tools and procedures, how they work, how they fall short, and what it would mean for them to be more representative, which will then enable the creation of a diagnostic tool to gauge effectiveness.

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Chapter 1

Introduction

In a social landscape where the public is growing progressively disillusioned with its representation in politics (Landemore 2026), it is important to consider ways in which we can foster more inclusive decision-making practices. One way of achieving this is using sortition to create panels that are more representative of the population for political discussions and deliberations. Adopted recently by an increasing number of institutions, including the Parisian ("Paris Creates a Permanent Citizens' Council" 2021), Irish (Walsh & Elkink 2021), and British Columbian (Ward 2008) governments, sortition as a part of democratic process is gaining traction. As a result, the need for fair and representative methods of building panels is only growing, and progress is being made in improving the structure of selective algorithms. However, there are still many challenges to overcome in order to ensure that these panels consistently achieve the level of representation that they envision. And there remains little procedure on measuring representativeness beyond the "surface" marginal level of demographics.

Sortition is a method of selection that uses random sampling to create panels that should be descriptively representative of a population. It has been used in various contexts, reaching as far back as ancient Athens (where it was how jurors were selected), and now into modern times for citizens' assemblies. The idea is that by randomly selecting individuals from the population, we can create panels that are more representative and inclusive than those selected through traditional methods, like elections or open-ended surveys. Due to its random nature, sortition also reduces dominance by elites or highly engaged groups, enabling broader participation across social classes (Giuliani 2025), making it more likely that decisions are in the relevant population's best interest.

One of the main roadblocks in sortition is ensuring that the selection process is transparent and fair, and that the panels are truly representative of the population in terms of demographics and perspectives. But this raises the questions of what it means to be fair, what it means to be representative, and how

these look from a systematic perspective.

As a process, sortition can be broken down into four stages: 1) defining the population and traits of interest, 2) requesting expressions of interest from a random subset of the population, 3) feeding respondents into a sortition algorithm, and 4) panel selection and engagement for deliberation. Notably, there is currently no rigorous evaluation of representation achieved before a process is run. Additionally, each of these steps has the potential to introduce bias, impacting the representativeness of the panel. As such, it is crucial to measure how the level of representativeness (or *potential* representativeness) flows through each stage, and to understand how different factors may affect it.

Sortition methods have been evolving over time. Prior to algorithmic means, panel selection was carried out through simple self-selection (response to survey/advertising) and proportional sampling to meet panel size requirements. As sortition began to grow more robust, the Legacy techniques (Flanigan et al. 2021) started surfacing—these saw panels put together through simple choosing of respondents who met the most required "category" condition at any given moment, and would continue iterating until the required marginal counts were present in the selected panel. This led to situations of extreme selection likelihoods, if minorities existed—as members were selected indiscriminately of implied likelihood, meaning there were situations where an individual would potentially be selected 0% of the time. To counteract this, the more systematic algorithm "MaxiMin" was implemented (Baharav & Flanigan 2024); seeking not only to meet panel quotas, but to raise the lowest individual's selection likelihood, with the aim of eliminating the 0% selection case, in order to improve representativeness. This, however, now presented the risk of likelihood manipulation, as selection was based more directly on the combination of all category responses given by an individual, rather than pure counts needed for the panel to be filled. As a result, the inverse function "MiniMax" was employed to reduce extreme cases on the other end—eliminating the 100% selection chance case by minimising maximal selection likelihood of any given individual, providing an approach that was more resistant and robust to coercion, but simultaneously reintroducing the potential 0% likelihood problem. Though robustness itself lies beyond the scope of this paper's aims, it is important to note the existence of MiniMax as an approach to the sortition problem, as it impacted further algorithmic development.

More recently, and of direct relevance to the current study, further iterations of MiniMax and MaxiMin, in the forms of "LexiMin" (Flanigan et al. 2021) and "Goldilocks" (Baharav & Flanigan 2024), were established. These present more generally practical outcomes, with the former being used in industry at present. LexiMin takes the idea of MaxiMin and repeats it for every individual sequentially, maximising *all* individuals' likelihoods, rather than just those who sit at the absolute bottom. Thus, raising the

selection likelihood floor for the participant pool overall. It is, however, like MaxiMin, vulnerable to manipulation (which remains beyond the scope of this paper)—leading into the development of the Goldilocks algorithm solution. Goldilocks aims to simultaneously raise the selection likelihood floor (eliminating the 0% selection chance and improving representativeness), as well as lower the ceiling (eliminating the 100% selection case and making the process more robust against manipulation). A tool for Goldilocks' use is in development at the time of the writing of this paper, and is expected to be made publicly available in the near future. As such, it will not be used for the current study, but may present an opportunity for further research.

Notably, all of these algorithms only focus on optimising for maximal selection likelihoods as their sole objective functions. This means that in its discovery and construction of feasible panels, the existing LexiMin algorithm may be prone to distortion of intersections and intersectional association trends present within population distributions. This presents a massive risk for the validity of representation achieved by panels output by the algorithm–the extent and reasoning of which will be explored further in the paper.

Additionally, there is limited literature on the visualisations of the outputs of these algorithms. This presents a great opportunity to demistify LexiMin and sortition as a process, improving transparency for practitioners, facilitators, participants, and broader community stakeholders alike. The comparative visualisation methods and tools also allow a general comparison to be made between any selected group and any broader sample it is pulled from, with any algorithm or selection process. This could be massively useful as a mechanism for quick sanity checks of panel configurations, and as a way for practitioners to evaluate the validity of panels before engaging in deliberation.

The remainder of this thesis is structured as follows: Chapter 2 provides a literature review of the existing research on sortition and representation, and identifies gaps in the current knowledge. Chapter 3 outlines the methodology used to evaluate the representativeness of panels and to investigate the impact of unobserved traits and survey non-response. Chapter 4 presents the results of the analysis, and Chapter 5 discusses the implications of the findings and suggests directions for future research.

Chapter 2

Literature Review

2.1 Representation and Sortition

2.1.1 What does it mean to be representative?

The main aim of any and all selection methods within sortition is to create a panel that is fair and representative of the population. But what does this actually mean?

Fairness refers to the idea that the selection process should be transparent, unbiased, and inclusive. It should give everyone an equal chance of being selected, regardless of their background or demography, which presents a selection likelihood difference minimisation challenge. Systematically, this is typically achieved through introducing randomness to selection processes (Flanigan et al. 2021), as not doing so immediately introduces implicit selection bias.

Alongside this, representativeness refers to the idea that the selected panel should accurately reflect the demographics and perspectives of the population from which it was built. This means that the panel should be diverse and inclusive, and should contain individuals from a range of prediagnosed backgrounds of interest. This can be assessed directly by comparing demographic compositions of the selected panels against the population from which it was originally drawn.

Approaching it procedurally, the sortition process can be thought of in four stages:

- Defining the population and traits of interest
- Sampling from a random subset of the population (through survey or other outreach)
- Panel selection (through algorithmic means or other method)
- Panel evaluation, confirmation and engagement

Each of these steps presents an opportunity for bias to be introduced, and for the panel to become less representative of the population. Which is why it is crucial to evaluate the level of representativeness (or *potential* representativeness) at each stage, and to understand how different factors may affect it.

For example, if the population of interest and the investigated traits are poorly defined, then the selection process may not be able to capture the diversity and perspectives of the desired group, leading to a less representative panel. This presents a challenge in itself, given most population descriptions (as previously discussed) are often based on disjoint demographic characteristics and do not provide a way to capture intersectionality, which is a key aspect of representativeness. However, this may be beyond the scope of this thesis, as it is already a foray into broader data collection and storage methods, but it is important to keep in mind that representation goes beyond the margin.

Similarly, if certain groups are less likely to respond to the survey, then they may be underrepresented in the pool of potential panelists, which can lead to a less representative panel from the outset. Illustratively–if there were to be no female respondents for a certain survey in an area comprised of 60% women, then the representativeness assumption (and goal) would be majorly violated before an attempt at putting a panel together has even been made. Situations like these raise the question of how we can account for this non-response bias, and whether we can use visualisation tools to show the discrepancy between the pool and the population (which leads back to part of this thesis' aim).

2.2 Panel Selection

2.2.1 Problem Definition

Suppose there is a pool of individuals numbered 1 to n.

We want to choose a panel of size k.

Let K be the set of all feasible panels. A feasible panel is any group of k individuals that satisfies the required constraints (which generally take the form of quotas).

These quotas will be certain desired counts, or ranges of counts, of individuals belonging to a certain trait class (e.g. 20-22 males, or 15-17 rural). These traits and quotas, by extension, will have direct ties to the demographics that have been deemed crucial to represent for a certain project's population group. Tying back to the established background–the core goal of sortition, then, is to build a panel which meets the quota constraints, equitably, robustly, and in a way that represents the population from which it was drawn.

2.2.2 Existing Methods

Non-algorithmic/self-selection

This is the most basic method of panel selection, where individuals are simply allowed to self-select into the panel. This can be done through a variety of means, such as advertising the opportunity to participate in the panel and allowing individuals to sign up on a first-come, first-served basis. However, these types of methods are often not very effective at creating representative panels, as they can a) lead to self-selection bias (as involvement will be tied to exposure and not random chance, those who put their hand up of their own volition are more likely to have pre-formed perspectives, ahead of the panel process), thus possibly not capturing the diversity and perspectives of the population, and b) lead to lower-than-desired engagement, as it does not guarantee any particular level of exposure or direct engagement in the first place.

LEGACY heuristic (Flanigan et al. 2021; Sortition Foundation 2024) Already a systematic improvement over non-algorithmic methods; the LEGACY selection heuristic takes a given sample of respondents and selects individuals one-by-one until the panel is full. At each step, the heuristic calculates a "need" value to identify the trait that is currently furthest from the desired count in the panel. LEGACY then selects a random individual from the pool who possesses that trait. If selecting any given individual would cause the panel to hit the upper quota for any trait, then all other individuals with that trait are removed from the pool before the next selection. This process is repeated until it either becomes infeasible (exhausts the dataset before hitting overall quota, at which point it will restart), or builds a full panel with desired quota representation. LEGACY's process fulfils the base requirements of building a panel that meets the desired quotas, but it does not necessarily maximise representativeness, as it does not take into account the likelihood of selection for different individuals (allowing them to just be 0%, in extreme cases), and it does not attempt to minimise the maximum deviation from equal selection chance across all individuals. This heuristic was, for a time, the Sortition Foundation's own in-practice method of panel selection, as implemented in their GroupSelect tool, before the more recent algorithmic fairness methods below were developed and adopted.

MaxiMin (Flanigan, Kehne & Procaccia 2021) and MiniMax (Baharav & Flanigan 2024)

This aforementioned shortcoming is then addressed by our first algorithmic sorition methods: MaxiMin and MiniMax, though these both tackle the problem in opposite ways and with differing goals. MaxiMin, as the name would suggest, is a selection algorithm that generates potential panels and selection probabilities and then sets out to maximise the minimum selection probability of the set. This reduces the likelihood of an individual never being chosen (improves fairness) but makes the method quite susceptible to manipulation, if an individual elects as belonging to certain quota groups

with lesser representation in the sample. Thus, leading to potential representative burden. Inversely, MiniMax aims to minimise the maximum probability of an individual being selected, with the aim of making selection more robust (less vulnerable to manipulation). However, this then comes at the cost of fairness, as the issue of potential 0% selection likelihood returns.

LexiMin (Flanigan et al. 2021)

LexiMin, or lexicographic minimisation, is a method of panel selection similar to MaxiMin, that aims to maximise the minimum likelihood of selection, but for each individual being selected sequentially (rather than just the outright minimum). It works by first identifying the individual with the lowest selection chance, and then selecting panel combinations that make this individual's selection chance as high as possible. This process is repeated for each individual in order of their selection likelihood until overall selection chance is maximised and, ideally, made as equal as possible across all participants. This ensures that each individual has the highest possible chance of being selected. But this trait of LexiMin can, like MaxiMin, make it vulnerable to manipulation.

Goldilocks (Baharav & Flanigan 2024)

Bringing us to the supposed current gold standard: the aptly named "Goldilocks" sortition algorithm. Goldilocks is structured in such a way that it simultaneously attempts to maximise minimum selection probabilities (improve fairness) and minimise maximum selection probabilities (improve robustness), providing a "best of both worlds" solution to the problem by averaging outcomes. The motivation for this balancing act is specifically to address MaxiMin's vulnerability to manipulation: because MaxiMin places no limit on how high an individual's selection probability can climb, a participant (or coordinated group of participants) who misreports their trait membership to appear as part of an under-represented quota group can inflate their own chance of selection disproportionately. By simultaneously capping the maximum selection probability any individual can receive, in the manner of MiniMax, Goldilocks directly limits the payoff available to such manipulation, improving robustness, while still prioritising the lowest selection probabilities enough to preserve most of MaxiMin's fairness guarantees. In this sense, Goldilocks is motivated less by a desire for some abstract "best of both worlds" balance, and more by the concrete, practical need for a selection rule that is simultaneously difficult to game and reasonably fair.

2.2.3 LexiMin approach (Flanigan et al. 2021)

Rather than committing to a single panel immediately, LexiMin instead constructs a probability distribution over the full set of feasible panels K, assigning each feasible panel $S \in K$ a non-negative probability p_S such that the probabilities across all panels sum to one. From this distribution, an individual's overall chance of selection can be calculated by summing the probabilities of every feasible

panel that would include them. Sorting these individual selection probabilities from smallest to largest, the LexiMin rule then proceeds lexicographically: it first makes the very smallest selection probability as large as possible, then, subject to that smallest probability remaining at its now-maximised value, makes the second-smallest probability as large as possible, and continues in this way until every individual's selection probability has been addressed in turn.

2.2.4 Formal Representation

This process can be expressed formally. Let p_S denote the probability assigned to panel S, which must be non-negative and sum to one across every feasible panel in K,

$$p_S \geq 0, \qquad \sum_{S \in K} p_S = 1.$$

The resulting selection probability for an individual i is then the sum of the probabilities of every feasible panel containing them,

$$\pi_i(p) = \sum_{S \in K,\ i \in S} p_S,$$

giving the full vector of individual selection probabilities across the population,

$$\pi(p) = (\pi_1(p), \pi_2(p), \ldots, \pi_n(p)).$$

Sorting this vector from smallest to largest,

$$\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq \ldots \leq \pi_{(n)}(p),$$

LexiMin chooses the distribution p that maximises this sorted vector in lexicographic order – that is, the smallest entry first, then the second-smallest subject to the first remaining maximised, and so on down the sorted vector.

2.2.5 Algorithm

In practice, this lexicographic maximisation is carried out as a sequence of steps: first, every feasible panel in K is listed, along with every possible probability distribution over those panels. For each candidate distribution, the resulting selection probability $\pi_i(p)$ is computed for every individual i, and these are sorted from smallest to largest, $\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq \ldots \leq \pi_{(n)}(p)$, as above. Only

the distribution (or distributions) that maximise the smallest selection probability $\pi_{(1)}(p)$ are then retained; if more than one still remains, this is narrowed further to only those that also maximise the second-smallest selection probability $\pi_{(2)}(p)$, and so on, resolving ties one probability at a time until a single distribution is determined. A final panel is then sampled according to that distribution.

2.2.6 Logic

LexiMin is, at its core, a fairness rule: it gives priority to the individuals with the lowest chance of being selected, improving their chances first, before iteratively improving the next-lowest probabilities until no further improvement is possible without disadvantaging an already-prioritised individual. This makes the selection probabilities as equal as possible across the population, while still respecting the panel's quota constraints.

2.2.7 Visualisation Methods

In order to evaluate the representativeness of selected panels, we used a variety of visualisation methods to show how the distribution of traits in the selected panel compares to the sample from which it was pulled, and to identify any discrepancies or biases in the selection process. Some potentially useful visualisation methods include: bar charts, mosaic plots, double decker plots, UpSet plots and parallel coordinate plots.

First, it will be of use to just do some illustrative graphing to show densities within the distribution across traits, in the form of bar charts. These will give a rough visual idea of the breakdown of the population and samples within the simulation, setting up some expectations for how representation should look in further visualisations. From there, either mosaic or double decker plots will be useful for depicting the proportion of given demographic traits that are selected for the panels. Finally, UpSet plots (Conway, Lex & Gehlenborg 2017) will be an effective way to showcase the intersections that exist in generated panels, as they offer a visual link between demographic variables, as well as frequency counts to highlight how often these occur within a given panel.

Attribute distribution analysis

- Showcase how the distribution of traits in the selected panel compares to the overall population, and to identify any discrepancies or biases in the selection process.

2.2.8 Discussion

This approach to the problem of panel selection has been shown to be effective for ensuring maximally equal selection probabilities, which can help to make sortition as a process fairer (reducing situations where specific individuals are massively disadvantaged and less likely to be selected). However, it does not give a way to introduce specific joint trait requirements for subsets of the population, which

may be a necessity for accurate representation of community groups. Additionally, as a whole, this limitation can violate representativeness as a goal, considering the motivation for sortitioned panels in the first place–filling a room with individuals who accurately and descriptively represent the a community's population diversity and characteristics.

2.2.9 New Investigation

Across the literature surveyed above, the common thread is that algorithmic effort has gone almost entirely into optimising the fairness and robustness of the selection *mechanism* itself – how equal selection probabilities are, and how resistant they are to manipulation – rather than into validating the *outcome* that mechanism actually produces. Put differently, a panel satisfying all of a given algorithm's fairness and robustness guarantees is not thereby guaranteed to be a good representation of the population it was drawn from; these are related but distinct properties, and the literature has so far focused heavily on the former at the expense of the latter.

One specific way this gap manifests is that, across all algorithmic techniques, stratification for combinations of demographic traits is not included within their decision-making designs–focusing instead purely on affecting individual selection chances based on quotas of each demographic trait at the margin. This is a major limitation, as it means that the selection process may not be able to entirely capture the diversity and perspectives of the desired group, leading to a less representative panel. In some situations (e.g. supporting side-panels built for specific subgroups, communities where there are known intersectional minorities of interest), it may be beneficial, or even necessary, to ensure the inclusion of specific intersectional combinations of traits in order to truly represent a given population. Though this limitation is generally mitigated in practice by bringing in "specialist speakers" to talk to the panel during their deliberation processes (Redman & Carson n.d.), this still is not the same type of inclusion by nature, as being directly part of the panel and decision-making process.

The extent to which intersectional inclusion happens naturally through current methods was evaluated in the later simulations. We then also establish a means to measure the level of intersectionality and association preserved throughout the selection process.

Chapter 3

Methodology

3.1 Research Aims

This thesis aimed to assess whether current algorithmic methods are achieving representative panels, to diagnose whether association trends present in a sample are preserved through the algorithmic sortition process, to demystify the sortition selection process in order to improve its transparency, and to provide a framework through which practitioners can make more informed and standardised decisions.

Underlying these aims is a more basic question: what does it mean for a panel to be "good" at being representative, and are current methods actually achieving this? This breaks down into several more specific questions. Are current methods and algorithms, such as LexiMin, performing the best they could be? How well are intersectional associations and trends preserved throughout selection, and therefore how representative is panel composition? And how can differing levels of survey response and other sampling-level issues bias representativeness, beyond whatever the selection algorithm itself contributes?

3.2 Research Design

3.2.1 Simulation design

This thesis investigates the impacts of data complexity and construction on algorithmic performance. To test this, a population of simple intersectional complexity (two variables) and varying associations (independent and associated) was simulated. However, the findings are generalisable to cases with greater numbers of categories, and are expected to be more severe in these contexts (given the intersectional matrix would only grow in dimension, while the selected panel size would remain

constant). Categories used were inspired by demographic variables often used for sortition in industry. These were age (taking values of 18-34, 35-44, 45-54, 55-64, 65+) and tenure status (taking values of rent, own and other). The combinations of these variables were then used to produce the 4 simulated datasets for the panel selection process.

3.2.2 Population, slice, sample and panel: procedure and definitions

Every simulated dataset in this thesis moves through the same four stages, and the terms used for each are kept deliberately consistent throughout to avoid ambiguity, since "sample" in particular is used loosely (and inconsistently) in much of the sortition literature.

- **Population**: the full simulated universe of individuals ($n_{\text{total}} = 100{,}000$), generated from a specified joint probability array over the category variables. This represents the entire community eligible for selection, with known marginal and joint (intersectional) distributions, since these were specified directly in the data-generating process.
- **Slice**: a simple random sample, without replacement, of 10% of the population, drawn before any response behaviour is simulated. This represents the group of individuals actually invited to take part – e.g. mailed a survey or invitation – and is sometimes called the *invited pool* elsewhere. Because it is drawn uniformly at random from the population, the slice's own composition deviates from the population's only through ordinary sampling variability, not through any systematic bias.
- **Sample**: the subset of the slice who actually respond, determined by an independent Bernoulli response trial for each invited individual, under one of the response-rate cases described below. This is the *realised* pool of individuals who are actually available for panel selection, also referred to as the *respondent pool* or *survey sample* elsewhere in this thesis, and is the benchmark against which every panel's representativeness is compared, since it – not the full population – is what any practical algorithm has to draw from, within current practices.
- **Panel**: a fixed-size subset of k members drawn from the sample by a selection algorithm (LexiMin, in current practice, via Panelot) subject to quota constraints on the permitted count of each category level. This is the final body that would convene to deliberate, and is the unit whose representativeness this thesis is ultimately concerned with evaluating.

Each stage can introduce its own departure from the population's true composition: the slice through ordinary sampling variability, the sample through differential response rates, and the panel through the selection algorithm's own quota mechanism (**?@sec-panel-association-methods**). Distinguishing these stages explicitly makes it possible to attribute any observed distortion in a final panel to a specific stage of the pipeline, rather than to sortition "in general".

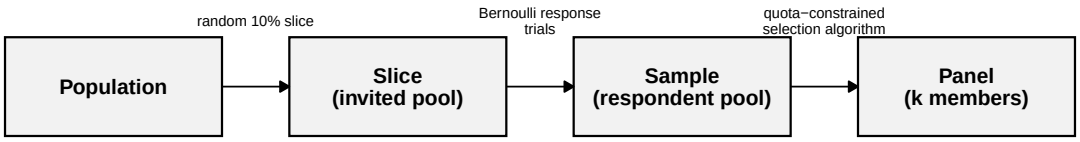


Figure 3.1: *The four stages of the simulation pipeline: population, slice, sample and panel.*

3.2.3 Two-variable illustrative example: age and tenure

To illustrate the general procedure concretely at a minimal scale, this thesis constructs a simple two-variable example, using only age and tenure. Age is denoted A, taking one of $L_A = 5$ ordered levels $a \in \{1, \ldots, 5\}$ (18-34 through 65+), as previously stated, and tenure is denoted T, taking one of $L_T = 3$ levels $t \in \{1, \ldots, 3\}$ (Rent, Own, Other). Marginal probability vectors were first defined for each variable,

$$\mathbf{p}_A = \left(p_{A_1}, \ldots, p_{A_5}\right), \qquad \mathbf{p}_T = \left(p_{T_1}, p_{T_2}, p_{T_3}\right), \qquad \sum_{a=1}^{5} p_{A_a} = 1, \qquad \sum_{t=1}^{3} p_{T_t} = 1,$$

with $\mathbf{p}_A = (0.15, 0.20, 0.25, 0.25, 0.15)$ specified as a mild skew toward the upper-middle age brackets, and $\mathbf{p}_T = (0.30, 0.50, 0.20)$ specified with a moderate skew toward home ownership. The same marginal targets $\mathbf{p}_A$ and $\mathbf{p}_T$ are used for both the independent and associated populations described below, so that any difference between them is attributable solely to the association introduced, rather than to a difference in either variable's marginal distribution.

An independent population (`sim_data2`) was generated by taking the outer product of these two marginals to form a joint probability array,

$$P^{(0)}(a,t) \;=\; p_{A_a}\, p_{T_t}, \qquad \mathbf{P}^{(0)} \;=\; \mathbf{p}_A \mathbf{p}_T^{\top} \;\in\; \mathbb{R}^{5\times 3},$$

a 5×3 matrix whose row sums equal $\mathbf{p}_A$ and column sums equal $\mathbf{p}_T$ by construction, so that $A \perp T$. This array was flattened into a probability vector and cell counts were drawn from a single multinomial trial before being expanded into individual-level records, as described in full below.

A second, associated population (`sim_data2_cov`) was then generated using the same marginal targets, but with a negative association deliberately introduced between age and renting. This required an ordinal score for age,

$$s(a) = \frac{a-1}{5-1} \in [0,1],$$

running from 0 for the youngest bracket to 1 for the oldest, which was used to seed the independent array $\mathbf{P}^{(0)}$ with an exponential adjustment factor targeting only the "Rent" level of tenure,

$$\tilde{P}(a,t) \;=\; P^{(0)}(a,t) \cdot \exp\Big(-\alpha\, s(a) \cdot \mathbb{1}[t = \text{Rent}]\Big), \qquad \alpha = 1.2,$$

so that the probability of renting is scaled down by a factor of $e^{-\alpha s(a)}$ that grows with age, while the probability of owning or other tenure is left unchanged. Because this adjustment is applied only to the "Rent" column, the row and column sums of $\tilde{\mathbf{P}}$ no longer equal $\mathbf{p}_A$ and $\mathbf{p}_T$, so the array was subsequently rescaled through iterative proportional fitting (raking) so that the age and tenure marginals were restored exactly to their specified targets. Initialising $P^{(0)}_{\text{rake}} = \tilde{P}$, each raking iteration $r = 0, 1, 2, \ldots$ performs two rescaling passes,

$$P^{(r+1/2)}(a,t) \;=\; P^{(r)}_{\text{rake}}(a,t) \cdot \frac{p_{A_a}}{\sum_{t'} P^{(r)}_{\text{rake}}(a,t')}, \qquad P^{(r+1)}_{\text{rake}}(a,t) \;=\; P^{(r+1/2)}(a,t) \cdot \frac{p_{T_t}}{\sum_{a'} P^{(r+1/2)}(a',t)},$$

first rescaling every row to sum to p_{A_a}, then every column to sum to p_{T_t}. Repeating this pair of passes for $R = 100$ iterations (well past convergence for a 5×3 table) yields the raked array $P^{\star}$, satisfying

$$\sum_t P^{\star}(a,t) = p_{A_a} \;\text{ for all } a, \qquad \sum_a P^{\star}(a,t) = p_{T_t} \;\text{ for all } t,$$

to numerical tolerance, while retaining the negative age-renting association introduced by the exponential seeding step – this is the sense in which raking restores the specified marginals while preserving the association structure introduced by the adjustment.

Both populations were sampled the same way, differing only in which array was used. The relevant array ($\mathbf{P}^{(0)}$ for the independent case, or the raked $P^\star$ for the associated case) was flattened into a probability vector $\mathbf{p}^\star \in \mathbb{R}^{15}$ (age varying fastest), and cell counts were drawn from a single multinomial trial,

$$(n_1, \ldots, n_{15}) \sim \text{Multinomial}(n_{\text{total}}, \mathbf{p}^\star), \qquad n_{\text{total}} = 100{,}000,$$

with each cell j, corresponding to one (a, t) combination, expanded into n_j individual-level records sharing that age and tenure. Comparing the two resulting populations demonstrates the core logic of the simulation framework in its simplest form: a chi-squared test of association between age and tenure is non-significant in the independent population but highly significant in the associated population, with the conditional probability of renting falling steadily from around 44% among 18-34 year olds to around 18% among those aged 65 and over, even though both populations share identical marginal distributions for age and tenure individually. This confirms that the seeding-and-raking procedure successfully introduces a targeted association between variables without disturbing their overall marginal proportions. This will allow for testing of the algorithm's ability to preserve sample intersection structures, beyond a purely marginal level.

Next, as is standard in practice, a random slice of defined proportions (in this case 10%) is taken from the population, to be used as the survey sample, from which responses are drawn. Concretely, from a population of n_{total} individuals, a survey sample $\mathscr{S}$ of size $n_{\mathscr{S}} = \lfloor 0.10\, n_{\text{total}} \rceil$ (so $n_{\mathscr{S}} = 10{,}000$ for the two-variable example, drawn uniformly at random without replacement) is taken as the pool of individuals invited to respond, before any response behaviour is simulated. Whether each invited individual $i \in \mathscr{S}$ actually responds is then modelled as an independent Bernoulli trial,

$$R_i \overset{\text{ind.}}{\sim} \text{Bernoulli}(\pi_i),$$

where the response probability π_i is assigned according to one of two cases: an independent case, and an age-covaried case mirroring the age-renting association introduced at the population level above.

In the *independent* case, every invited individual is equally likely to respond regardless of their trait profile,

$$\pi_i = \pi_0, \qquad \pi_0 = 0.30,$$

providing a baseline against which the effects of differential response can be compared – a chi-squared test confirms no significant association between A and R under this case, as expected. In the *age-covaried* case, response likelihood increases with age, reusing the same ordinal score $s(a)$ defined above,

$$\pi_i = \pi_1 \exp\big(\alpha_R\, s(a_i)\big), \qquad \pi_1 = 0.20, \quad \alpha_R = 1.0,$$

so that the response rate rises from around 20% among the youngest bracket to over 50% among the oldest, and the association between A and R is now highly significant.

3.2.4 From survey sample to panel

Applying the independent and age-covaried response cases to both the independent and associated populations produces four respondent-pool files – `sim_data2_ind_eq.csv`, `sim_data2_ind_ass.csv`, `sim_data2_cov_eq.csv` and `sim_data2_cov_ass.csv` – each containing only the individuals who actually responded under that population/response-case combination, together with their age and tenure. These file names follow the same `<ind|cov>_<eq|ass>` convention used for the lottery files below, so that a file name alone identifies both the population and response case it was drawn from. Each of these respondent pools is, in the terminology introduced above, a *sample*: the realised pool from which a panel can actually be drawn. Each respondent-pool file was then passed to Panelot's LexiMin algorithm, along with quota constraints on the permitted count of each age bracket and tenure category (based on the population's marginal distributions, with ±1 individual slack allowed for the 18-34 and 65+ brackets, as they would require non-integer solutions otherwise), to enumerate every feasible panel of $k = 50$ members, yielding four corresponding lottery files (`lottery_ind_eq.csv`, `lottery_ind_ass.csv`, `lottery_cov_eq.csv` and `lottery_cov_ass.csv`) that are evaluated in **?@sec-panel-association-methods**.

3.2.5 Validating the simulation: numerical summary

Alongside the mosaic plots used throughout this chapter to visually compare the age x tenure population, slice and respondent sample, each simulated dataset is also checked numerically, to confirm that the data-generating process behaved as intended before it is used for any downstream

panel evaluation. Two quantities recur throughout this numerical checking, and are used again, in a more formal role, in the panel-level association testing described in **?@sec-panel-association-methods**:

- A **chi-squared test of independence**, applied to the 5×3 age-by-tenure contingency table, is used to confirm that the intended association (or lack of it) was actually realised in a given dataset. A non-significant result ($p > 0.05$) for the independent population (`sim_data2`), alongside a significant result for the associated population (`sim_data2_cov`), is the expected pattern and serves as a sanity check on the simulation code itself, rather than a substantive finding in its own right.
- Marginal **proportion tables** (`prop.table()` applied to the age and tenure frequency tables) are used to confirm that each variable's marginal distribution matches its specified target ($\mathbf{p}_A$ or $\mathbf{p}_T$) to within sampling error, at both the population and sample level. This is particularly important for the associated population, where the raking procedure is specifically designed to restore exact marginal targets despite the association-inducing adjustment applied beforehand.

Later, when panels themselves are evaluated (**?@sec-panel-association-methods**), the same chi-squared logic is extended into a three-way log-linear homogeneity test, and supplemented with Cramer's V as an effect-size measure, so that a panel's divergence from its sample can be judged by both its statistical significance and its practical magnitude, rather than significance alone.

3.2.6 Method limitations and caveats

The response rates used for the age x tenure pipeline – a 30% baseline in the independent case, rising to over 50% in the age-covaried case – are considerably higher than response rates typically reported for real-world sortition processes, where response to a civic lottery invitation is often below 10%. This was a deliberate choice rather than an oversight: response rates this low, applied to a 10,000-person slice, would in some simulated conditions leave too few respondents in individual age-by-tenure cells (of which there are 15) to reliably assess an algorithm's ability to satisfy quota constraints or preserve intersectional structure at a panel size of $k = 50$, since the algorithm would be data-starved regardless of its own effectiveness. Elevating the response rate ensured sufficient data richness to isolate the algorithm's own behaviour as the object of study. A consequence of this choice is that any representativeness issues identified in this thesis should be read as a conservative, best-case estimate: at the lower response rates typical of practice, the same mechanisms (differential non-response, quota-constrained selection leaving the joint age-tenure distribution unconstrained) would be expected to have less data to work with and correspondingly more room to produce poorly representative panels, not less.

Additionally, every association deliberately introduced in the age x tenure pipeline – the negative age-renting association at the population level ($\alpha = 1.2$), and the age-response association at the sampling level ($\alpha_R = 1.0$) – was controlled by a single exponential strength parameter, fixed at a moderate-to-strong value chosen to make each association clearly visible in a mosaic plot and reliably detectable by a chi-squared test at this thesis's sample sizes. These particular values were not derived from any specific real-world dataset, and the extent to which the selection algorithm's representativeness or association-preservation performance degrades is unlikely to scale uniformly with the strength of the underlying association: a weaker real-world age-tenure association might be preserved adequately by the same selection process that fails to preserve the stronger one simulated here, or, conversely, a subtle association could be *more* easily disrupted by quota-constrained selection if it is based on a smaller number of affected individuals. Because this thesis's simulation uses a small number of fixed association strengths, rather than a systematic sweep across a range of values, the panel-level results in **?@sec-panel-association-methods** should be read as demonstrating that association-preservation failures of this kind are possible and can be severe at the strength simulated, rather than as a precise, generalisable estimate of their magnitude across the full range of association strengths that might occur in real-world sortition applications.

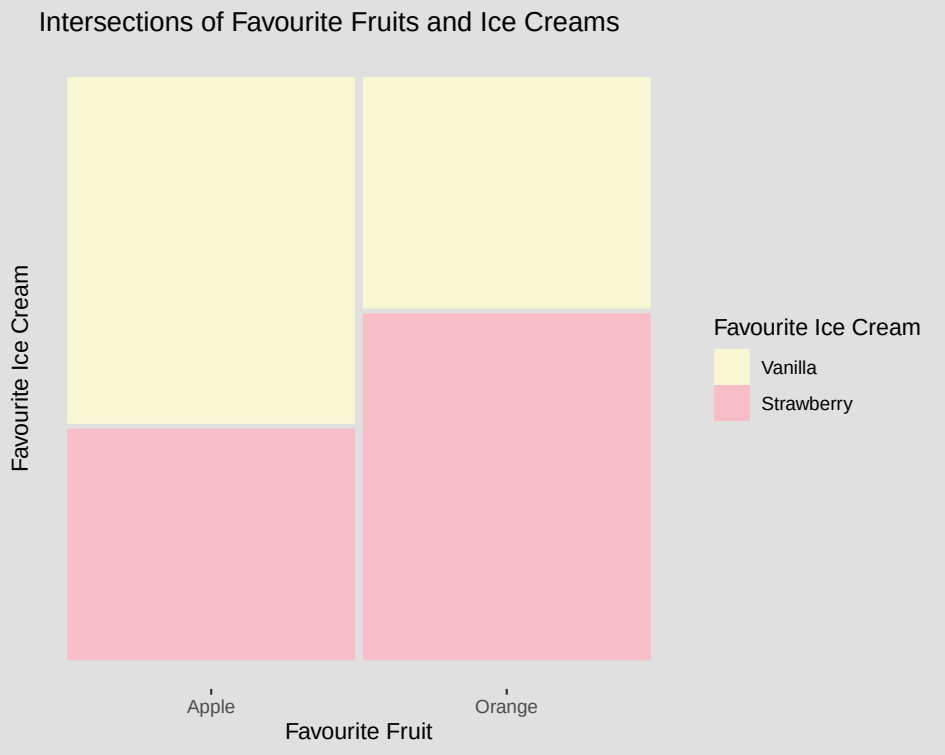
3.2.7 Visualisation methods

The choice of visualisation used throughout this thesis follows directly from the nature of the data being examined. The variables of interest – age brackets, tenure status, and the other demographic categories used for sortition – are categorical rather than continuous, and the central question is how these variables intersect, rather than how any one of them is distributed in isolation. Many standard visualisation tools are a poor fit for this combination of properties: scatterplots and histograms are built for continuous data; a simple bar chart or pie chart can show a single categorical variable's marginal distribution, but not how two or more categorical variables co-occur; and a bar chart faceted or grouped by a second variable quickly becomes difficult to read once cell counts become uneven (precisely the situation this thesis investigates) because bar heights are not naturally comparable to each other in the way that proportional tile areas are. As such, we landed on the mosaic plot, which, by contrast, is specifically designed to display exactly this kind of cross-classified categorical structure, representing both marginal distributions and their joint/intersectional distribution within a single figure. As such, they are the primary visualisation tool used throughout this thesis.

A mosaic plot is a way of visualising the joint distribution of two or more categorical variables, in which each combination of categories is represented by a rectangular tile whose area is proportional to the number (or proportion) of observations sharing that combination. This makes mosaic plots well suited to exactly the kind of question this thesis is concerned with: not just whether individual

categories are represented in the right proportions on their own (a question a simple bar chart could already answer), but whether particular *combinations* of categories – intersections – are over- or under-represented relative to what would be expected for their population demographies, and whether the variables are independent. Because tile area scales with both variables at once, a mosaic plot can reveal an association between categorical variables (a departure from independence) that would not be visible from either variable's marginal distribution alone.

To ensure later plots are easily interpreted, we provide the simplest form of a mosaic plot below:



For this plot, only two variables are examined: favourite fruit and favourite ice cream flavour. For the visualised example data set, 25 individuals with a preference for each marginal option were generated (for a total n of 50), then shuffled randomly to make independent intersectional combinations. As such, the width of each top-level bar is proportional to that category's marginal share of the data (here, the number of individuals preferring each fruit), while the height of the coloured segments stacked within a bar is proportional to the conditional proportion of the fill variable within that category (here, the proportion preferring each ice cream flavour, given a preference for that fruit). Because fruit and ice cream preference were shuffled independently of one another in this example, the two fruit bars are split at close to the same width, visually confirming that the two variables carry no association: knowing someone's favourite fruit provides no information about their favourite ice cream flavour. Where the fill proportions instead differ noticeably from one bar to the next, this indicates an association between the two variables – precisely the pattern that distinguishes the

independent and associated age-tenure populations introduced above.

Throughout the rest of this thesis, this same convention – categories on the x-axis with width proportional to marginal share, and a second variable shown via fill colour with segment height proportional to conditional proportion – is used for the core population, slice and sample comparisons, with age brackets on the x-axis and tenure as fill.

Chapter 4

Results

4.1 Simulation visualisations

4.1.1 Simulation pipeline: population, slice and respondents (equal-chance case)

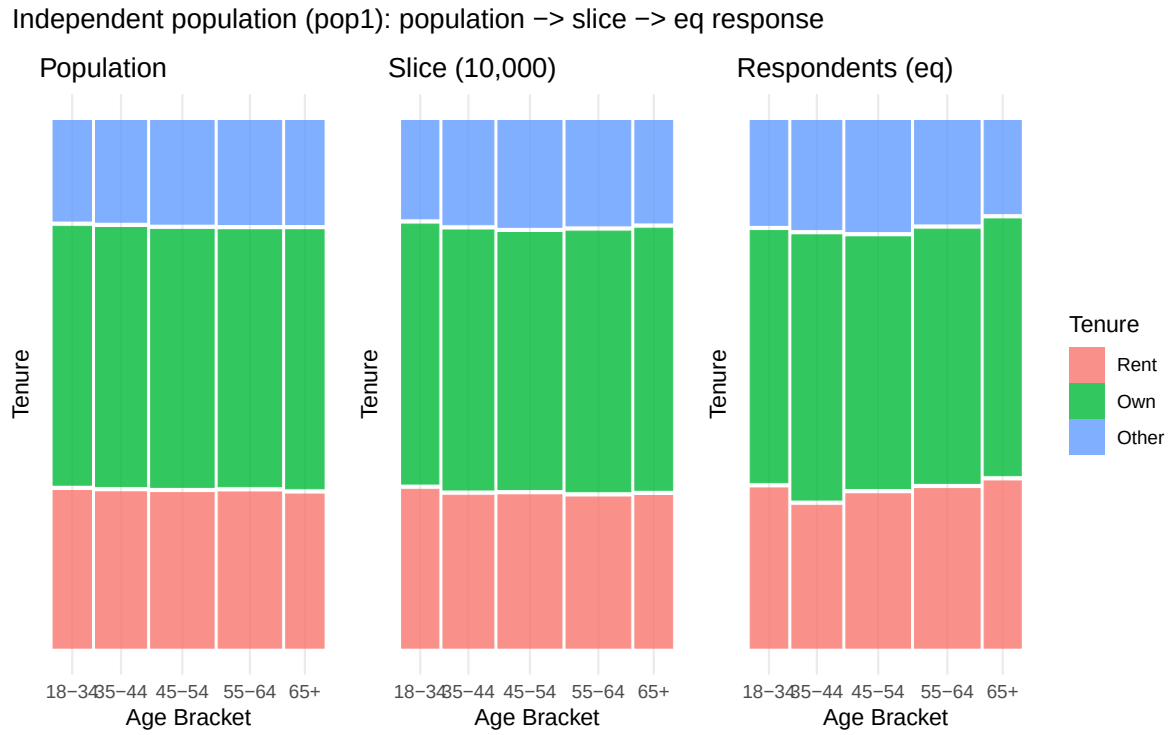


Figure 4.1: *Patchwork of mosaic plots tracing the independent population's age-tenure structure across the simulation pipeline: the full population, a random 10,000-person slice, and the slice's respondents under the equal-chance response case. Age brackets are shown on the x-axis and tenure as fill throughout.*

Figure 4.1 and Figure 4.2 trace each population's age-tenure structure across the three pipeline stages under the equal-chance response case. Both populations skew towards upper-middle age

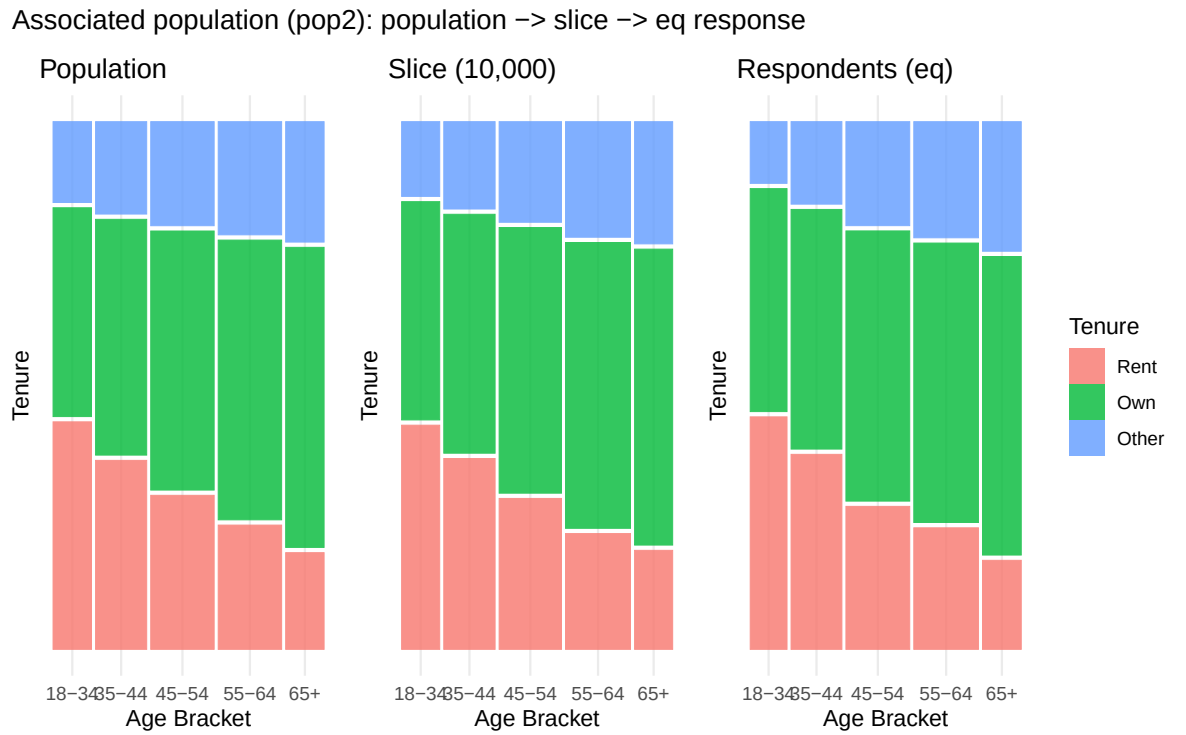


Figure 4.2: *Patchwork of mosaic plots tracing the associated population's age-tenure structure across the simulation pipeline: the full population, a random 10,000-person slice, and the slice's respondents under the equal-chance response case. Age brackets are shown on the x-axis and tenure as fill throughout.*

and a higher frequency of homeownership compared to other categories. In Figure 4.1, there is no visible association between age and tenure at any stage – the probability of any age-tenure intersection is explained purely by the margins – and this independence, along with each category's relative proportions, is preserved essentially unchanged from the full population through the 10,000-person slice to the equal-chance respondents. In Figure 4.2, by contrast, a negative association between age and renting is visible at every stage: the proportion of renters among the youngest bracket is noticeably higher than among the oldest. This pipeline also shows its association and relative proportions carrying through from population to slice to equal-chance respondents largely unchanged, confirming that neither random slicing nor equal-chance response meaningfully disturbs either population's underlying age-tenure structure.

4.1.2 Simulation pipeline: population, slice and respondents (age-associated case)

Figure 4.3 and Figure 4.4 repeat the same three-stage pipeline, but with response likelihood now rising with age rather than being equal across brackets. Unlike the equal-chance case in Figure 4.1 and Figure 4.2, the age-tenure structure is no longer preserved unchanged once respondents are reached: in both Figure 4.3 and Figure 4.4, the older age brackets visibly widen and the younger brackets narrow between the "Slice (10,000)" and "Respondents (ass)" panels, since older individuals

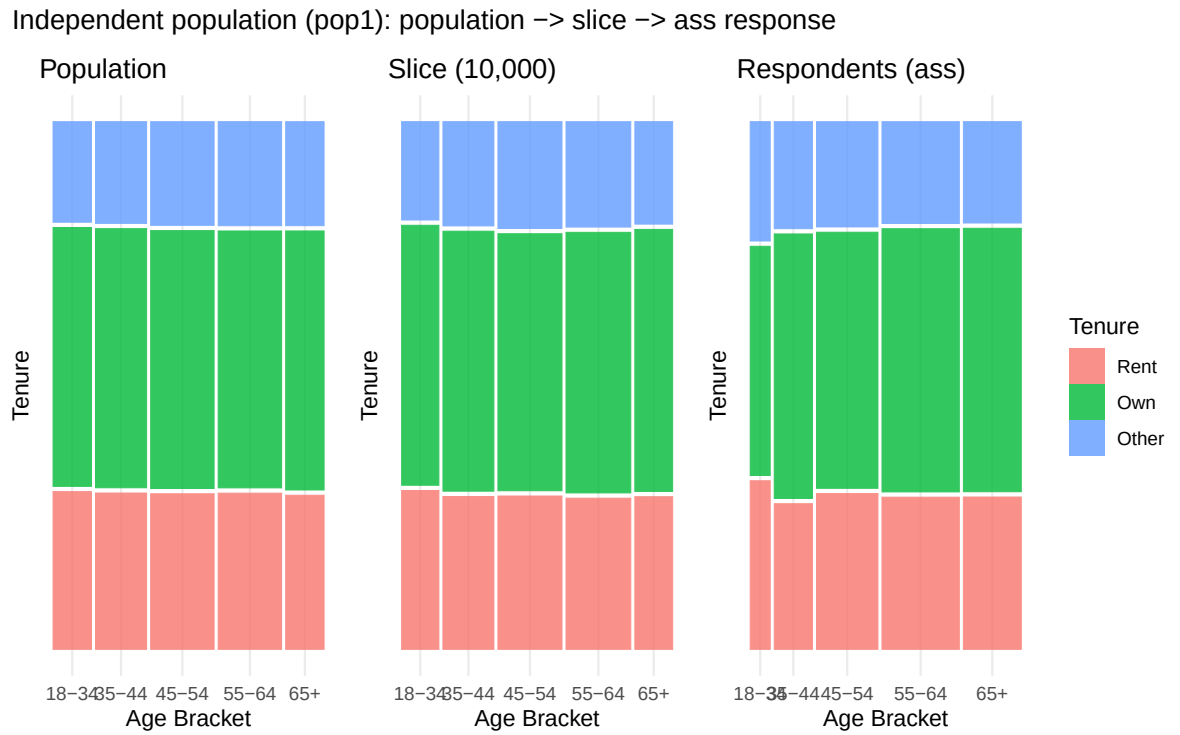


Figure 4.3: *Patchwork of mosaic plots tracing the independent population's age-tenure structure across the simulation pipeline: the full population, a random 10,000-person slice, and the slice's respondents under the age-associated response case. Age brackets are shown on the x-axis and tenure as fill throughout.*

are now disproportionately more likely to respond. This demonstrates that an association between age and response probability – entirely independent of any association between age and tenure in the underlying population – is sufficient on its own to shift the overall age composition, and therefore the intersections present, in the final respondent sample. Comparing Figure 4.4 specifically against Figure 4.2 shows this age-skew compounding with the associated population's existing negative age-renting relationship, rather than replacing it: the respondents under age-associated response are weighted even further towards the older, lower-renting end of the distribution than either the equal-chance respondents or the slice itself.

4.1.3 Quantifying association across the pipeline

The mosaic patchworks above show the age-tenure association visually; Table 4.1 quantifies it at every stage, using a chi-squared test of independence and Cramer's V as an effect-size measure of how strong that association is.

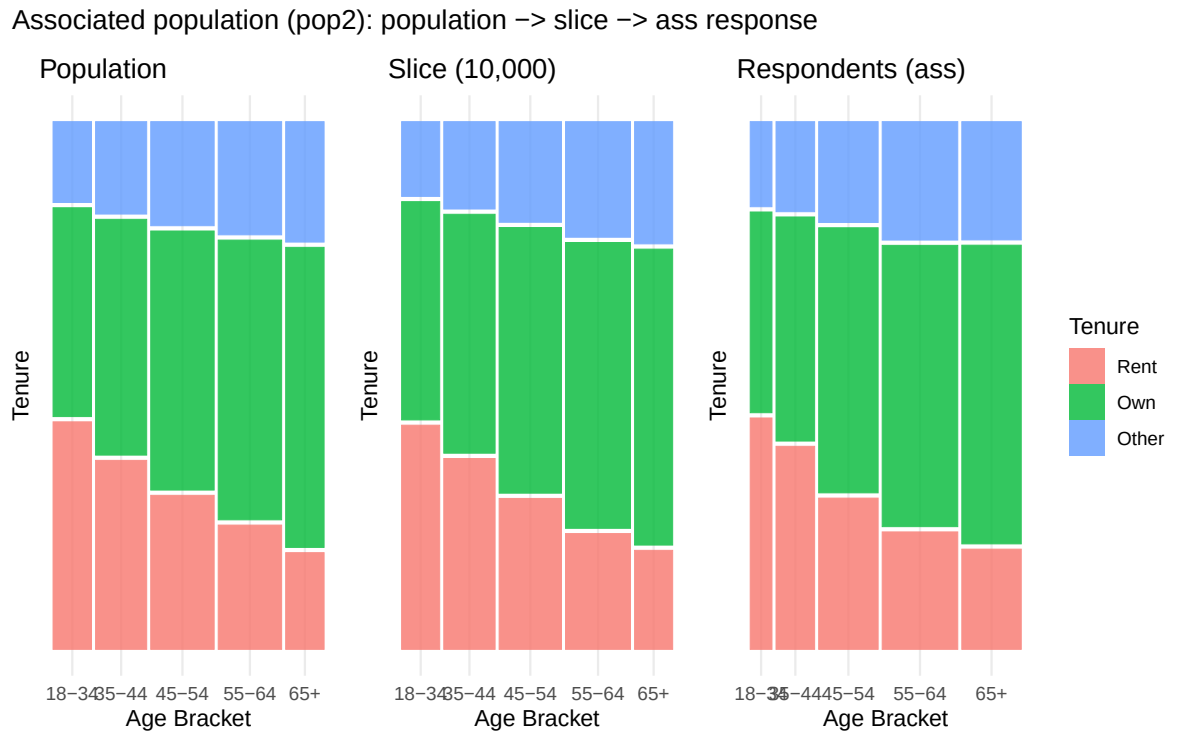


Figure 4.4: *Patchwork of mosaic plots tracing the associated population's age-tenure structure across the simulation pipeline: the full population, a random 10,000-person slice, and the slice's respondents under the age-associated response case. Age brackets are shown on the x-axis and tenure as fill throughout.*

Table 4.1: *Chi-squared test of independence and Cramer's V for the age-tenure association at each stage of the two-variable pipeline, for both populations and both response cases.*

Population	Response case	Stage	n	Chi-sq	df	p-value	Cramer's V
Independent (pop1)	–	Population	100000	5.2	8	0.735	0.005
Independent (pop1)	–	Slice (10,000)	10000	2.3	8	0.969	0.011
Independent (pop1)	eq	Respondents	2974	4.6	8	0.803	0.028
Independent (pop1)	ass	Respondents	3483	5.0	8	0.759	0.027
Associated (pop2)	–	Population	100000	3148.2	8	< .001	0.125
Associated (pop2)	–	Slice (10,000)	10000	334.3	8	< .001	0.129
Associated (pop2)	eq	Respondents	3019	123.9	8	< .001	0.143
Associated (pop2)	ass	Respondents	3563	123.4	8	< .001	0.132

Table 4.1 confirms what Figure 4.1 through Figure 4.4 show visually. For the independent population, no stage or response case ever reaches significance (all $p > 0.7$), and Cramer's V stays below 0.03 throughout – consistent with an age-tenure association that is genuinely absent, not merely undetected. For the associated population, every stage is significant at $p < .001$, and Cramer's V stays in a comparatively narrow band (0.125-0.143) from population through to both respondent cases, confirming that the underlying age-renting association is neither diluted by slicing nor meaningfully inflated by either response case – the age-associated case's effect is visible mainly as a shift in the respondents' age *composition* (Figure 4.3, Figure 4.4), rather than as a change in the strength of the age-tenure relationship itself. One caveat applies to the independent population's p-values specifically: with population and slice sample sizes in the tens of thousands, even this chi-squared test has the power to detect very small deviations from independence, so the consistently non-significant results here reflect a genuine absence of association, rather than a lack of statistical power.

4.2 Panel construction

4.2.1 Association preservation

It is found that 100% of panels have a significantly different association level than exists

4.3 Framework and recommendations

Chapter 5

Introducing Intersection Inspector

Having flagged the issues present with current sortitioning methods and process, the second gap sought to be addressed was the absence of a tool for practitioners to assess the viability and intersectionality of panels before engaging in deliberative processes. As such, a Shiny app was developed. Even within current practice, the LexiMin lottery and panel selection process took two steps: 1. uploading categories and trait quotas alongside a respondent pool and determining desired panel size, and 2. taking the computed panel lottery to a interactive page for random number generation to select a specific panel. The Shiny app developed as part of this thesis serves to replace this second step, with additional functionality that will benefit practitioners in smoothness of their workflow, as well as provide more confidence in the representativeness of their panels at a controllably granular level of intersectionality. This will enable practitioners to make more informed decisions about potential quota constraint adjustments and representative validity, minimising likelihood of unexpectedly poor intersectional spread and representation when panels convene.

5.1 Context: prior steps in the sortition pipeline

Intersection Inspector is not intended to replace the sortition process as a whole, only its final step. By the point a practitioner opens the app, they are assumed to have already carried out the preceding stages of a sortition: recruiting and surveying a respondent pool, specifying the category variables and trait quotas the panel should satisfy (e.g. minimum and maximum counts per age bracket, per tenure category), and running a selection algorithm over that pool to discover every panel of the desired size that satisfies those quotas. In current practice this selection step is performed by the algorithm LexiMin, through an app hosted at Panelot.org, which outputs a lottery file enumerating each feasible panel together with its probability of being returned.

The app takes this lottery file and the underlying respondent pool as given, and does not re-run or second-guess the selection algorithm that produced them. This also means its usefulness is not contingent on LexiMin specifically: any sortition algorithm or process that produces an equivalent output – a set of feasible panels (or a single panel) drawn from a known respondent pool, satisfying some quota specification – could be inspected through the app in the same way. The quota-based selection step and the intersectional evaluation step addressed here are conceptually separate, and the latter is agnostic to which algorithm was used for the former.

5.2 Intended workflow

The app page consists of three parts: 1. a sidebar comprised of interactable buttons, toggles and sliders, 2. a window that displays mosaic plots comparing the 2-way intersections and associations present between the sample and any selected panel for any two selected category variables, and 3. a window that displays the relative proportions of every trait present in the sample and any selected panel, across all categories present within the sample's dataset. To make use of this, a practitioner is expected to follow the below procedure.

5.2.1 Uploading files

The app, Intersection Inspector, takes two files as input: a lottery file (Panelot's `lottery.csv`, enumerating every panel the sortition algorithm found feasible, each with its member IDs and return probability) and the sample file the lottery was drawn from (the respondent pool that was invited and agreed to participate). From these two files alone, the app reconstructs every selected panel's composition on each category present in the sample and makes its intersectional structure directly inspectable, without requiring the practitioner to write any code or leave the browser.

5.2.2 Comparing intersections and marginal proportions

Once the files are uploaded, the practitioner selects any two category variables of interest (e.g. age and tenure) and the app renders a pair of mosaic plots side by side: one for the currently selected panel, one for the sample it was drawn from. These mosaic plots visualise the two-way intersection and association between the chosen variables, so that a practitioner can see at a glance whether the panel's joint distribution of traits resembles the sample's "shape", or whether it is skewed in a way that marginal quota counts alone would not reveal. Beneath the mosaic plots, a second window displays bar charts of every trait's marginal proportion in the panel against the sample, across every category in the dataset, covering representativeness at the single-variable level as well.

5.2.3 Testing and browsing panels

Beyond visual inspection, the app formally tests association preservation for each panel using the association tests described in the Chapter 4, and summarises each as a p-value and significance interpretation ("level of difference" from the sample), alongside the panel's size and its probability of being selected by the lottery. Panels can be browsed in several ways: sliding across all discovered panels ordered from most to least similar in association to the sample, jumping to a uniformly random panel (allowing arbitrary evaluation of the algorithm's performance), jumping to a probability-weighted random panel (sidestepping the need for a separate external random number generator once a panel is to be finalised), or entering a specific panel ID directly. A further view lets the practitioner check, at an adjustable representation threshold, which intersections of categories are missing or under-represented in the panel relative to the sample. Once a panel is chosen, its member IDs can be inspected and copied to the clipboard to continue the existing workflow (e.g. contacting selected members).

5.2.4 Replacing the Panelot.org selection step

In current practice, after a lottery file is generated, a practitioner takes it to a second, separate interactive page – on Panelot.org – purely to generate a random number and select one specific panel from those enumerated, then has to manually select the member IDs from the output. That step reveals no information about *which* panel is being selected or how representative it is beyond the quota constraints already satisfied; it is, in effect, a black box that turns a lottery file into a single panel. Intersection Inspector is designed as a drop-in replacement for this step: it accepts the same lottery and sample files, supports the same random (and probability-weighted random) selection as a final action, and returns the same member list a practitioner needs to proceed – while additionally providing the mosaic plots, marginal bar charts, association p-values, and under-representation checks along the way.

5.2.5 Benefits for new practitioners

This is particularly intended to benefit practitioners who are new to sortition or do not have a background in statistics. Rather than trusting that a tool with satisfied quota constraints is automatically representative, or treating panel selection as an opaque final step, the practitioner is given an interpretable visual and numerical basis for that trust (or for flagging concern) before a panel is convened. Because panels can be scrolled through and compared on their association preservation, a practitioner can also use the app diagnostically: if every available panel shows a similarly poor association to the sample, this points to the quota constraints themselves needing revision, rather than to any single unlucky draw. This also provides the practitioner an opportunity to decide whether any particular

intersections need to be captured for their particular engagement context, and determine whether additional surveying may be necessary for a more representative sample. Assuming the practitioner has access to the slice data, they can also format this to match the expected survey-respondent file structure and compare the panel against this directly, enabling a more direct comparison against data that would be closer to the population. In this way, the app turns a previously invisible step of the sortition pipeline into a transparent, inspectable one, directly in the hands of the people already responsible for running it.

App functions:

- Takes a lottery file and sample file
- Generates mosaic plots to show intersections of categories within lottery and sample
- Generates bar charts of trait proportions to show the relative marginal presence of traits between the panel and sample
- Calculates the level of similarity of associations/general shape present within the sample and panels
- Orders the panels based on their level of similarity in association to the sample, by using the calculated p-values
- Displays size of sample and panel, as well as selection probability of each panel and their level of association preservation compared to sample (as p-value, with interpretation of "significance level of difference")
- Allows the user to slide across these to compare as they worsen
- Allows the user to jump to a fully random panel, just to assess its level of representation
- Allows the user to input a specific panel ID, if they want to inspect it in particular
- Allows the user to jump to probability-weighted random panels (if probability exists within the lottery file), side-stepping the need for external RNG to select a panel
- Allows the user to inspect and copy panel member IDs to clipboard, to continue with workflow
- Allows the user to check missing and under-represented intersections (compared to sample) at different representation thresholds

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5.3 App function draft

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